

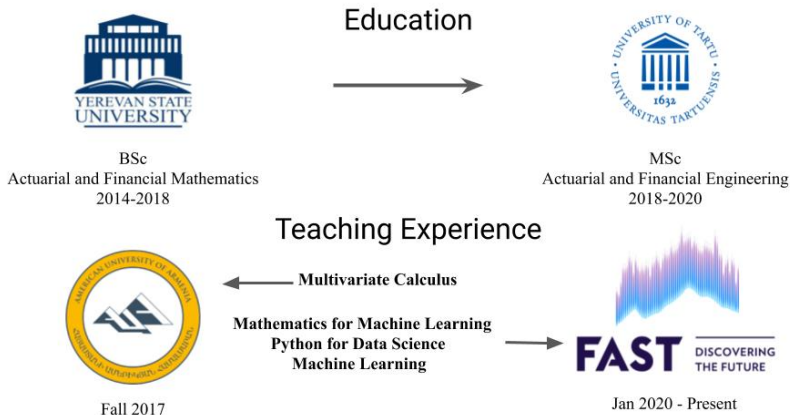
Machine Learning

Ingredients of Machine Learning

FAST

DISCOVERING
THE FUTURE

About Me (Nshan Potikyan)



- Lectures provide the theoretical knowledge

Course Format

- Lectures provide the theoretical knowledge
- Practice sessions support understanding of the lectures and provide practical knowledge

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- The practical sessions will be conducted by **Norayr Galikyan** and **Vahram Ustabashyan**

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Learning Objectives

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- **write the scripts of ML algorithms from scratch**

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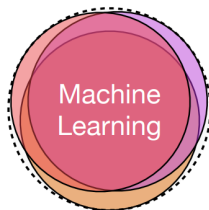
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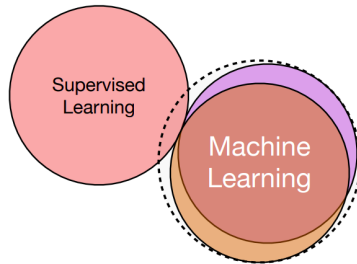
Mathematical Derivations

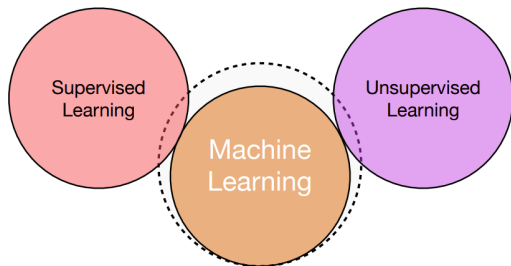
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- To fulfill the previous point **you need to ask questions**

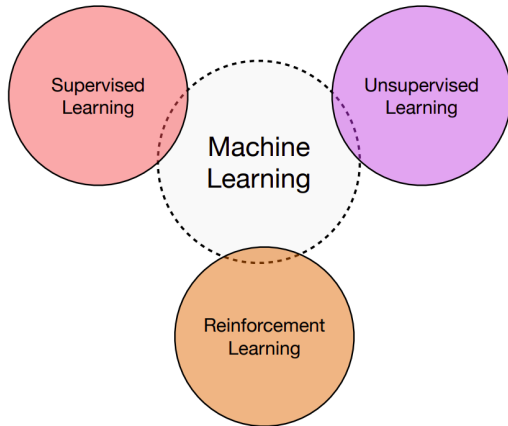
Topics of today's lecture

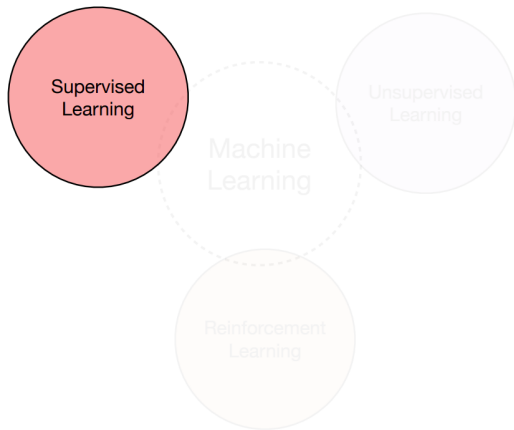
- Ingredients of Machine Learning
- Classification Basics
- Basic Linear Classifier

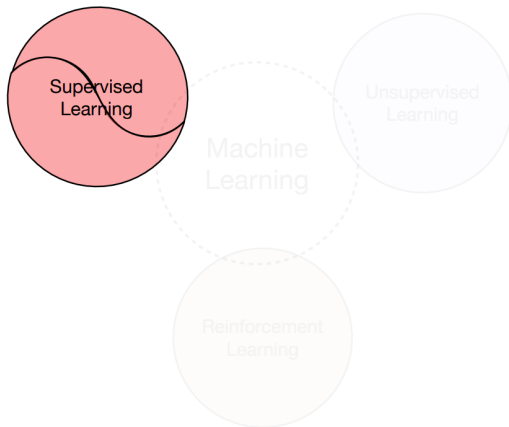


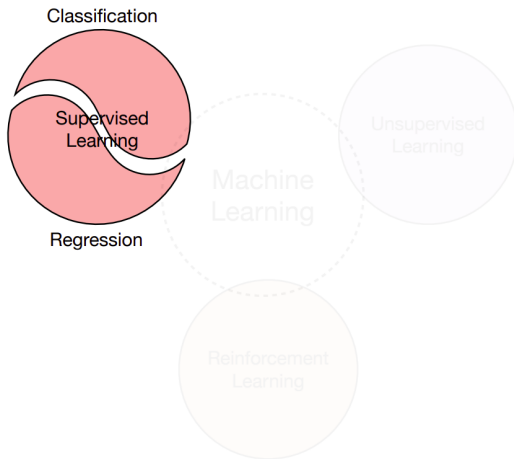


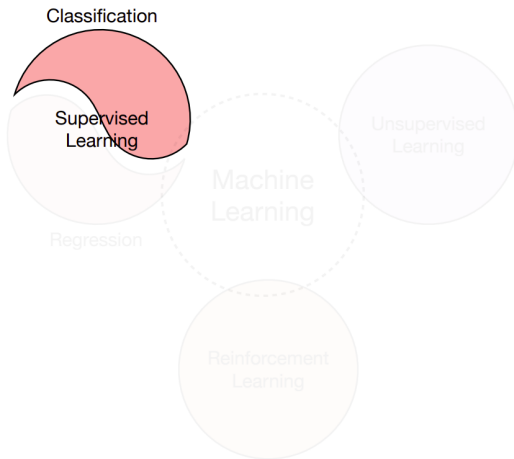


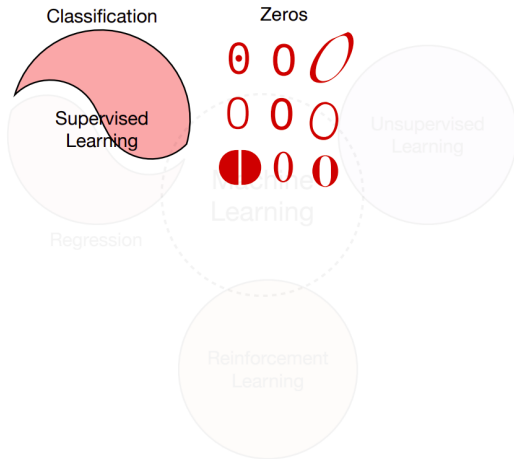


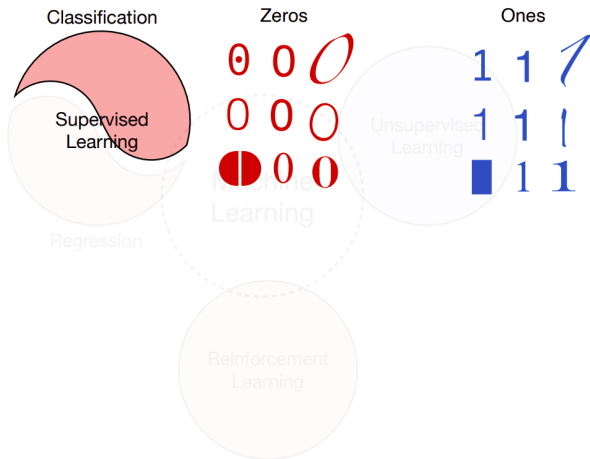


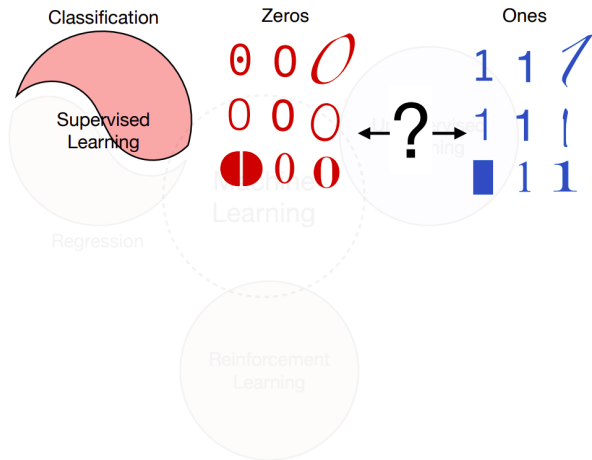




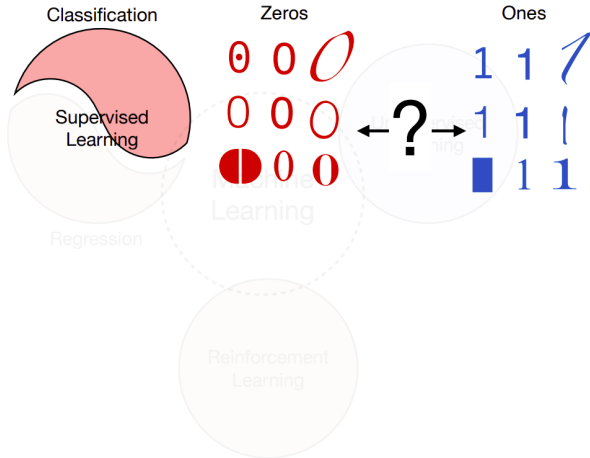




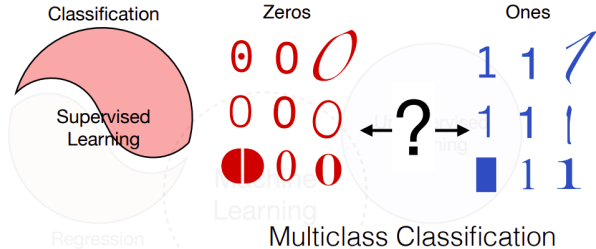




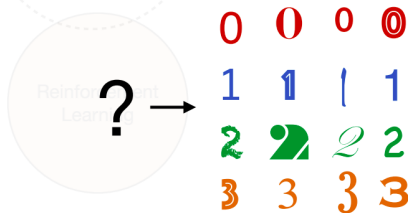
Binary Classification



Binary Classification

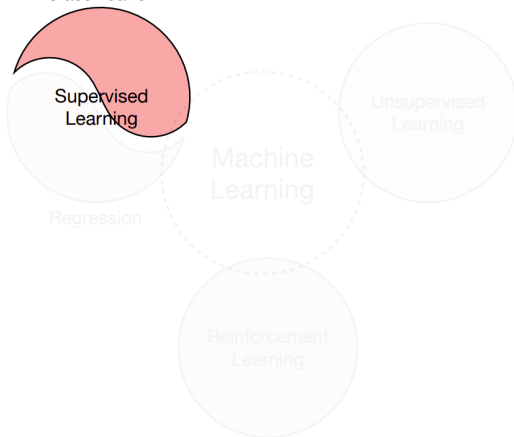


Multiclass Classification

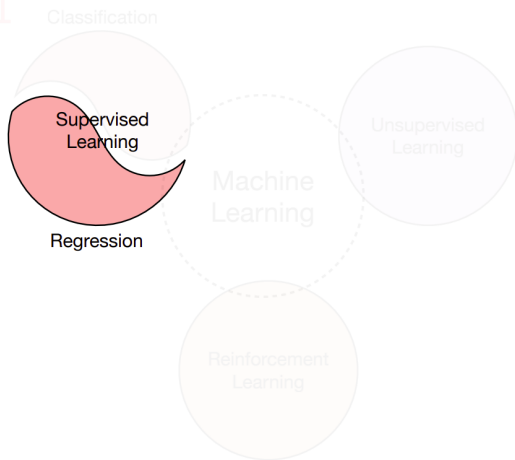


0 ← ? → 1

Classification



0 $\longleftrightarrow$? $\longrightarrow$ 1



0 $\longleftrightarrow$? $\longrightarrow$ 1

Classification

Supervised
Learning

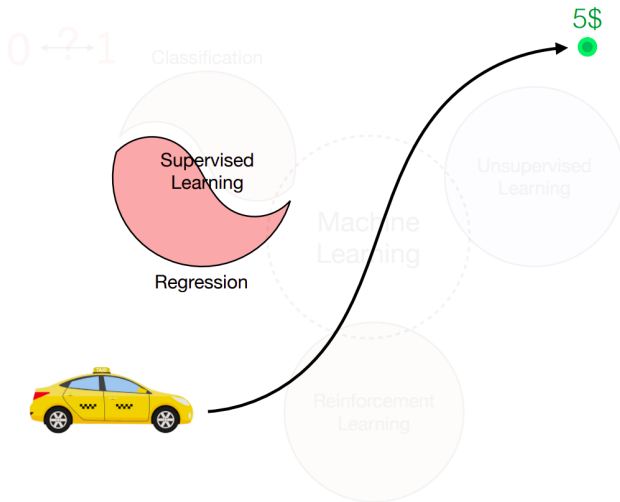
Regression

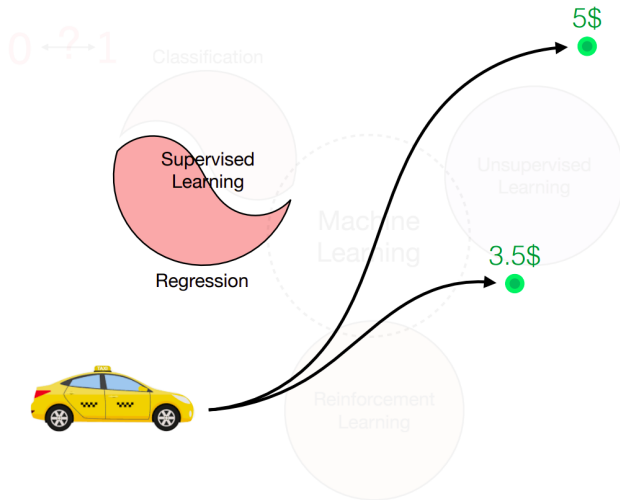
Machine
Learning

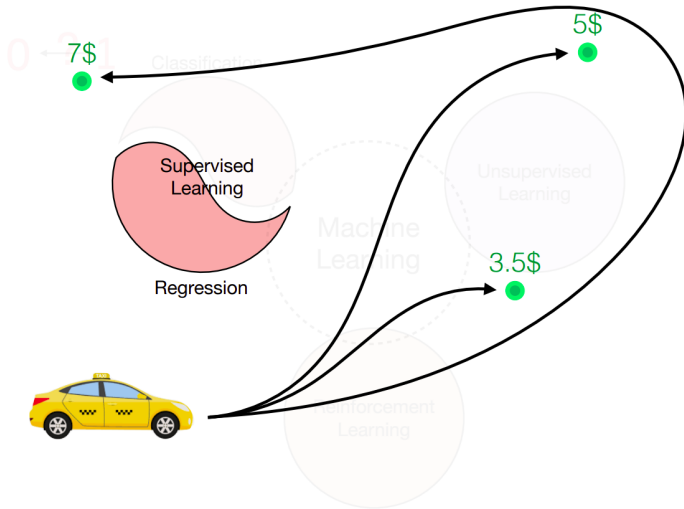
Unsupervised
Learning

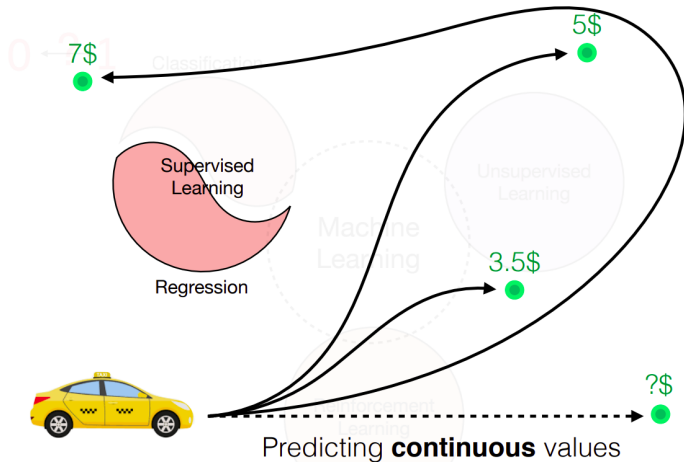
Reinforcement
Learning

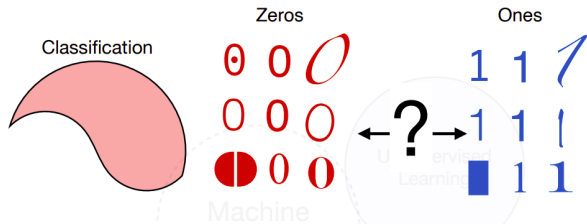










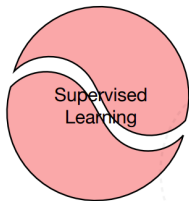


What is the main similarity/difference between these two classes of **supervised learning**?



$0 \leftarrow ? \rightarrow 1$

Classification



Regression

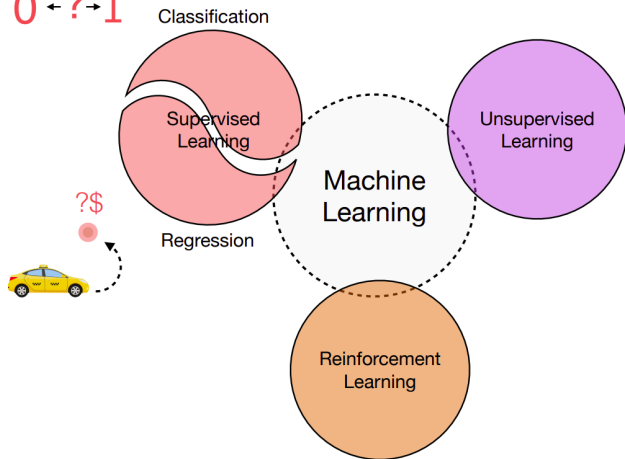


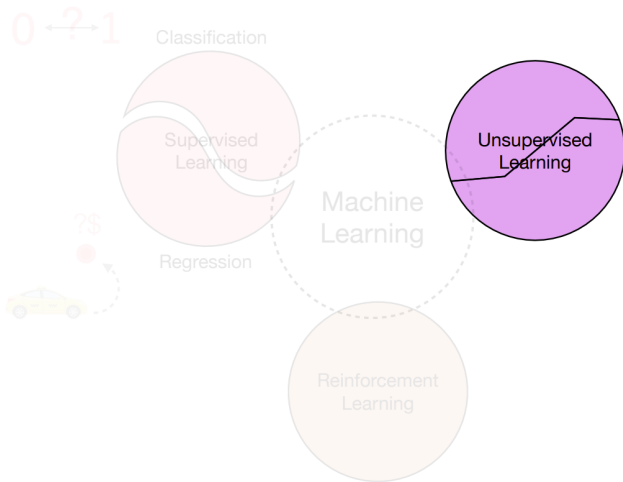
Machine Learning

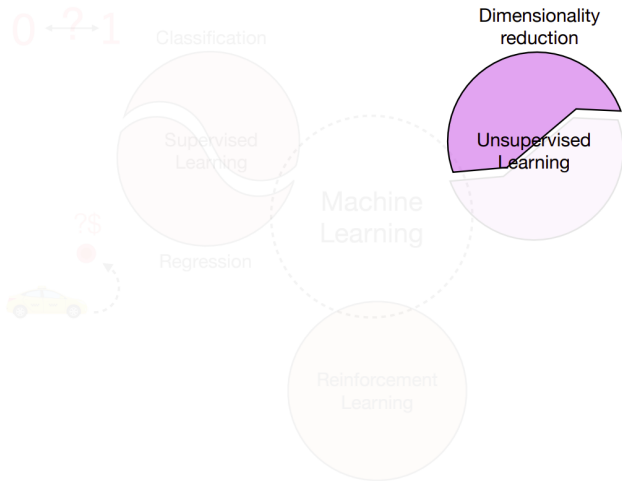
Unsupervised Learning

Reinforcement Learning

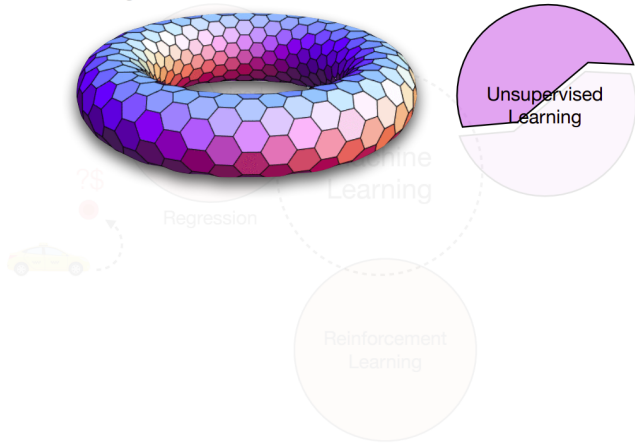
0 ← ? → 1



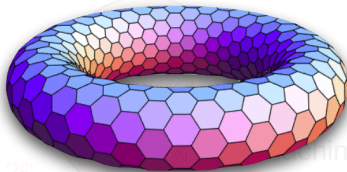




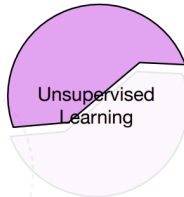
0 ← ? **High Dimensional Space**



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Dimensionality
reduction

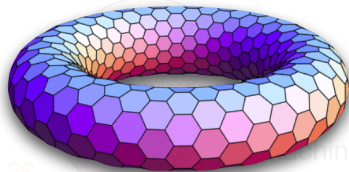


Dimensionality
reduction

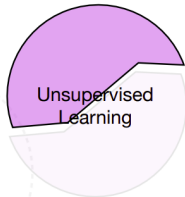


Reinforcement
Learning

0 ← ? **High** Dimensional Space



Dimensionality reduction

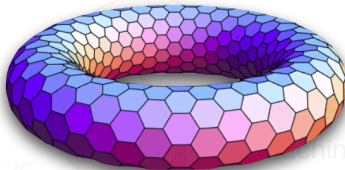


Dimensionality reduction

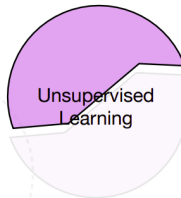


Low Dimensional Space

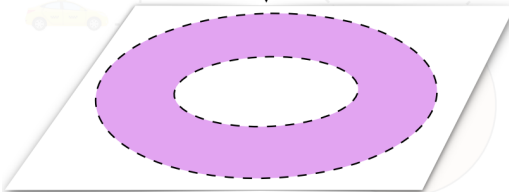
0 ← ? \$
High Dimensional Space



Dimensionality
reduction



Dimensionality
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Low Dimensional Space

0 $\longleftrightarrow$? $\rightarrow$ 1

Classification

Supervised
Learning

Regression

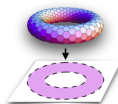


Machine
Learning

Reinforcement
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Dimensionality
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Unsupervised
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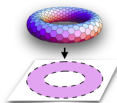
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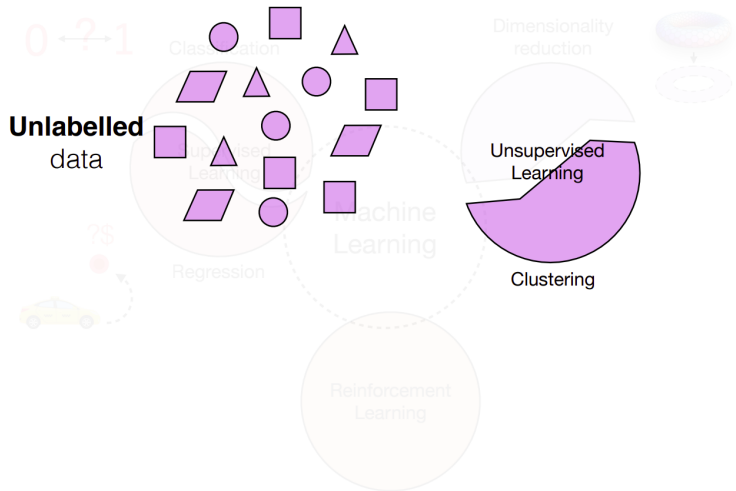
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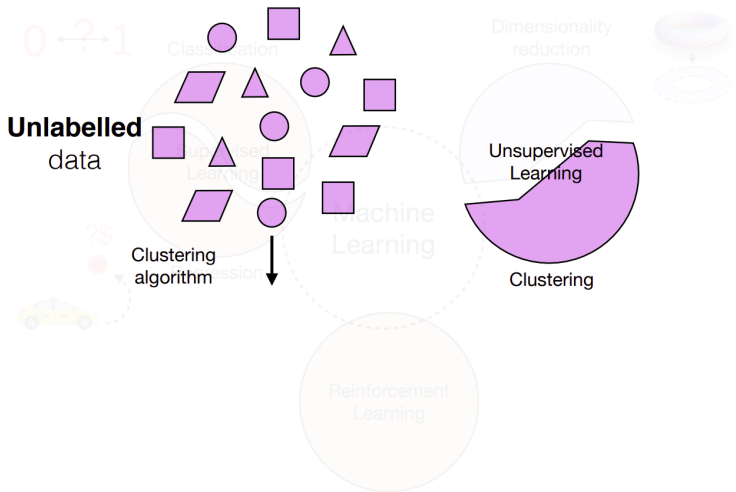
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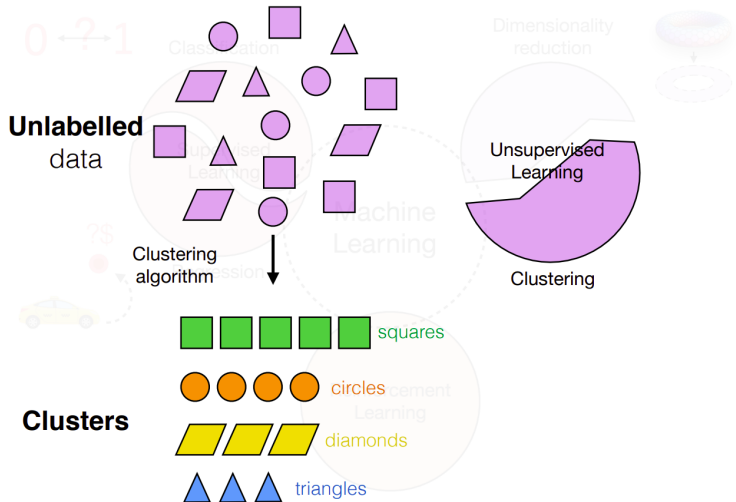
Unsupervised
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Clustering









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Regression



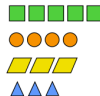
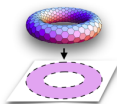
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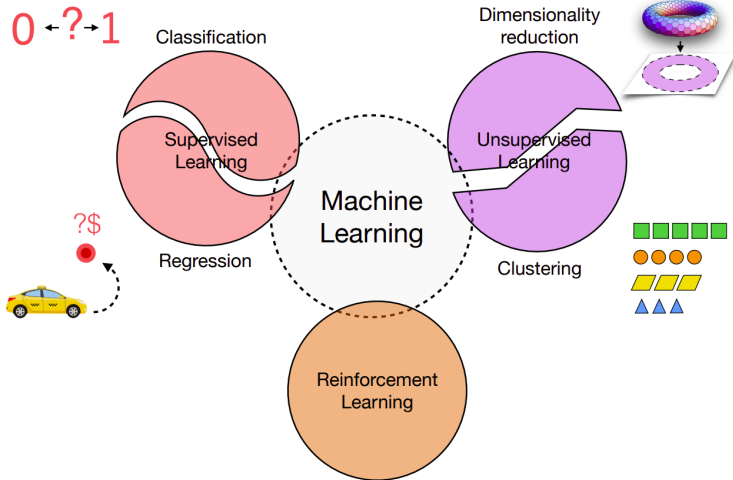
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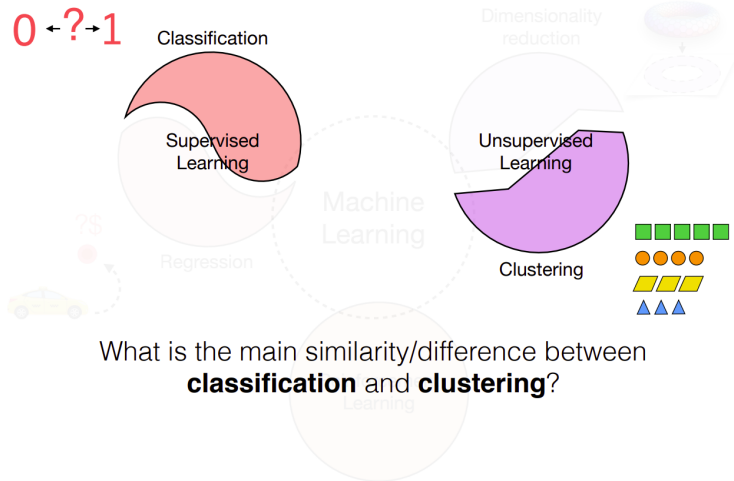
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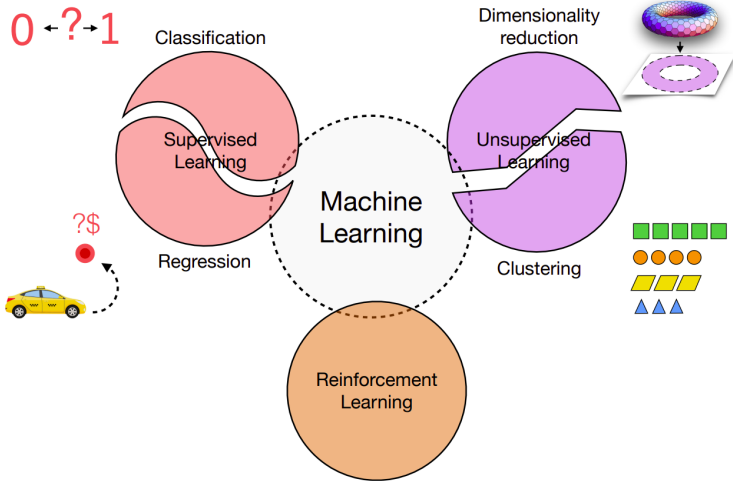
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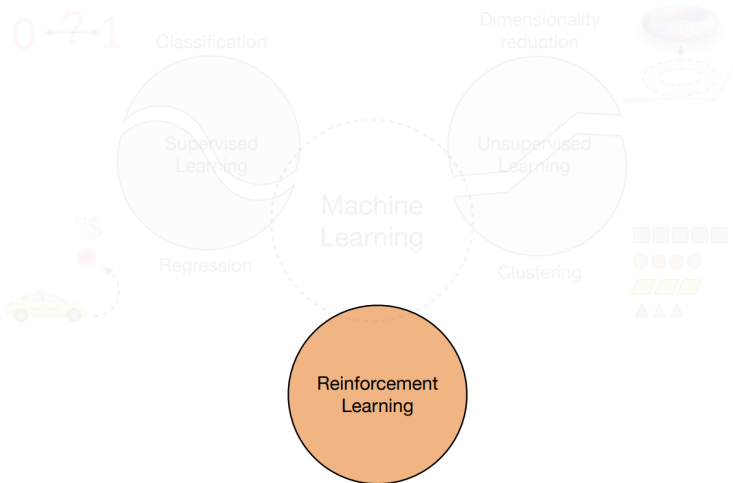


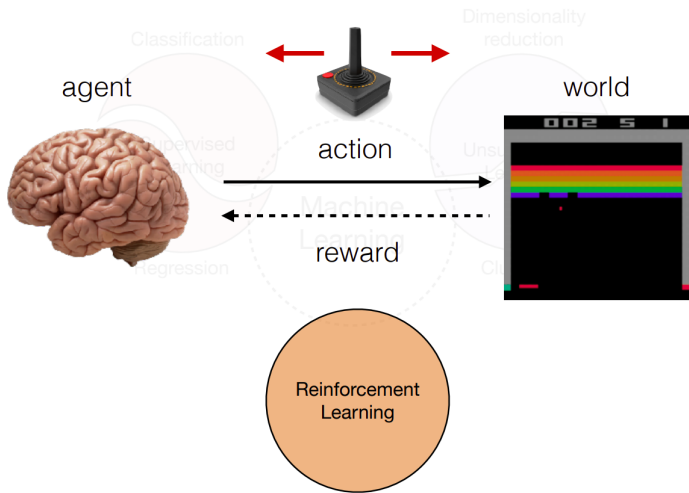
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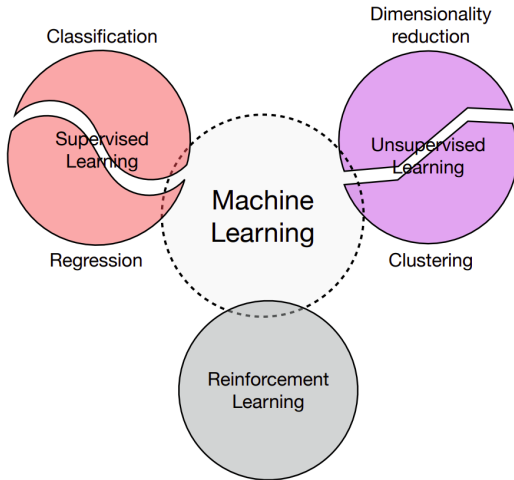


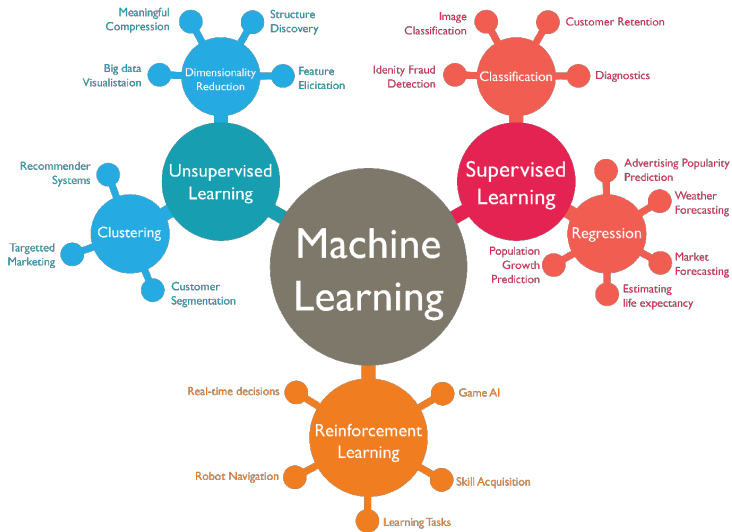


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Course Syllabus

- K-Nearest Neighbours

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- Naive Bayes

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- K-Means, K-Medoids, DBSCAN, Hierarchical Clustering
- Principal Component Analysis, t-SNE
- **Probabilistic Reasoning (Bayesian Networks, HMM, Kalman Filter, Particle Filter)**

Spam e-mail detector



Machine Learning Terminology



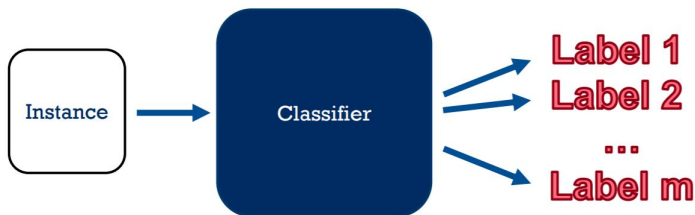
Machine Learning Terminology

- Binary classifier (spam e-mail detector)
 - Inputs an instance (e-mail)
 - Predicts positive or negative label (spam/ham)

Examples of binary classification

- Instance = photo, Label = cat / dog
- Instance = audio, Label = guitar / ukulele
- Instance = text, Label = English / Armenian

Classification



Here m is the number of different labels:

- Binary classifier: $m = 2$
- Multi-class classifier: $m \geq 2$

Mathematical notation of classification

- Notations can vary in the literature
- $\mathbb{X}$ - input space (set of all possible instances)
- $\mathbb{Y}$ - output space (all possible labels)
- $f : \mathbb{X} \rightarrow \mathbb{Y}$ - any such function is a classifier
- $\mathbf{x} \in \mathbb{X}$ - instance
- $y \in \mathbb{Y}$ - actual / true label of instance $\mathbf{x}$
- $\hat{y} = f(\mathbf{x})$ - predicted label of instance $\mathbf{x}$

- In classification we assume that every instance has an actual / true label y
- A classifier is perfect if it is always correct, i.e. it predicts the actual label: $\hat{y} = y$
- Usually classifiers make some errors: $\hat{y} \neq y$

How to build a spam detector?



- Let us think of how to build a spam detector
- Without machine learning for now

Building a spam detector

- What is a typical spam e-mail?

Building a spam detector

- What is a typical spam e-mail?
 - Advertising something that we don't want

Building a spam detector

- What is a typical spam e-mail?
 - Advertising something that we don't want
 - Trying to steal our money

Building a spam detector

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 - Trying to infect our computer with a virus
 - ...

Building a spam detector

- What is a typical spam e-mail?
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- What helps us to discriminate between spam and ham?

Building a spam detector

- What is a typical spam e-mail?
 - Advertising something that we don't want
 - Trying to steal our money
 - Trying to infect our computer with a virus
 - ...
- What helps us to discriminate between spam and ham?
- We need to find features that occur more often in spam than in ham (or vice versa)

100 binary features for spam detection

1. Does the e-mail contain an inline gif-image?
(yes / no)

100 binary features for spam detection

1. Does the e-mail contain an inline gif-image?
(yes / no)
2. Does the e-mail contain HTML with more images than text?
(yes / no)

100 binary features for spam detection

1. Does the e-mail contain an inline gif-image?
(yes / no)
2. Does the e-mail contain HTML with more images than text?
(yes / no)
3. Does the sender's hostname contain a long non-vowel sequence?
(yes / no)
- ...

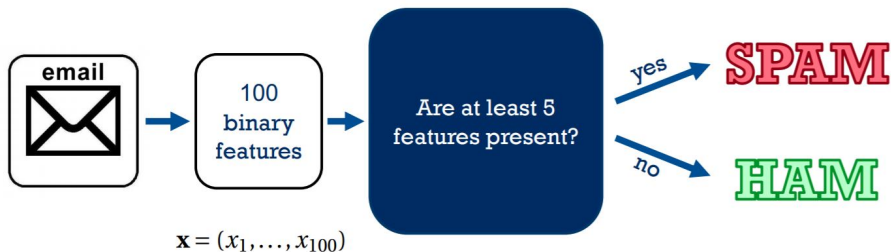
100 binary features for spam detection

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Our first spam detector

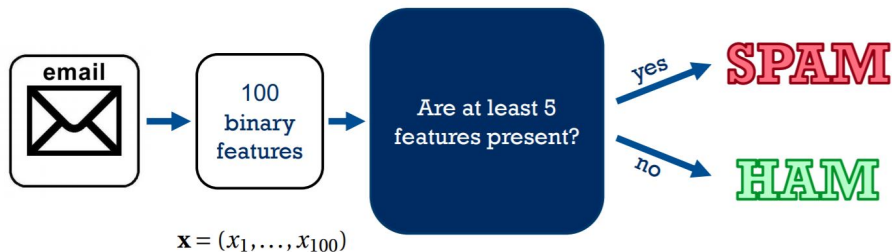
Classify as spam whenever at least 5 features are present.

Building a spam detector



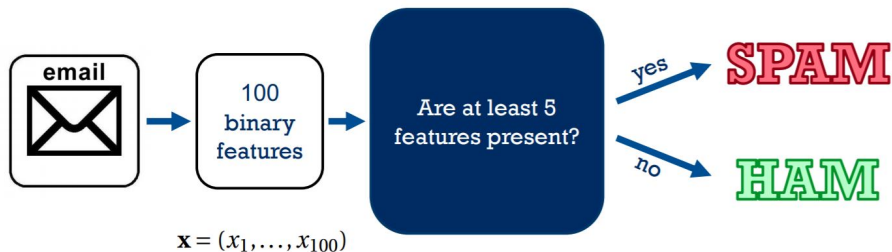
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Building a spam detector



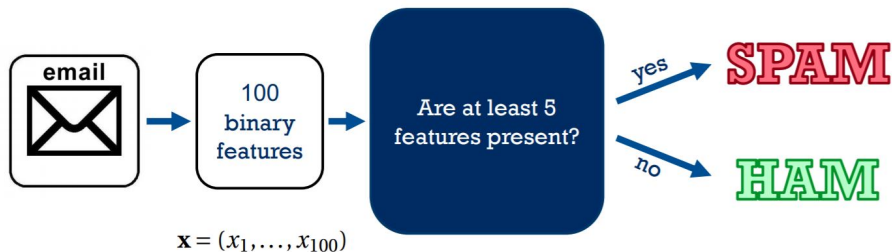
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Building a spam detector



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- Classifier: $\hat{y} = f(\mathbf{x}) = \begin{cases} \text{spam, if } x_1 + x_2 + \dots + x_{100} \geq 5 \\ \text{ham, otherwise} \end{cases}$

Building a spam detector



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- What if some features are more important than others?

Making some features more important

- Let's make
 - x_1 three times more important than it was
 - x_2 two times more important than it was
- Our new classifier:

$$\hat{y} = f(\mathbf{x}) = \begin{cases} \text{spam, if } w_1x_1 + w_2x_2 + \dots + w_{100}x_{100} \geq 5 \\ \text{ham, otherwise,} \end{cases}$$

where $w_1 = 3, w_2 = 2, w_3 = w_4 = \dots = w_{100} = 1$

- This particular classifier belongs to the family of **linear classifiers**

Definition

A binary classifier is called a **linear classifier** if it predicts one class whenever

$$\sum_{i=1}^m w_i x_i \geq t$$

and predicts the other class otherwise. Here $w_i \in \mathbb{R}$ for $i = 1, \dots, m$

Apache SpamAssassin ¹

- Anti-spam platform giving system administrators a filter to classify email and block spam
- Initial release: Apr 2001
- Latest release: Apr 2021
- Still in active use
- Uses a linear classifier on top of constructed binary features

¹https://en.wikipedia.org/wiki/Apache_SpamAssassin 

How does SpamAssassin work?

- Takes an e-mail
- Runs several tests on it
- Calculates a score based on the weights of each test
- If the score is 5 or more then classifies as spam, otherwise as ham

How to choose good weight values?

- Take a big set of labelled e-mails (known as the **training set** in machine learning)
- Make sure that your linear classifier makes as few mistakes as possible on this set
 - That is, it predicts the correct label (spam/ham) for as many e-mails as possible
 - Ideally, correct on all e-mails
- We could try through a huge set of possible weight combinations and find the best combination

Example

- We run 2 tests (x_1, x_2)
- We have 4 training e-mails
 - 1 spam (label: Spam=1)
 - 3 ham (label: Spam=0)

E-mail	x_1	x_2	Spam?
1	1	1	1
2	0	0	0
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- Which weights would be good with threshold 5?

Which weights would be good?

- Let's denote weights by w_1, w_2

E-mail	x_1	x_2	Spam?	$w_1x_1 + w_2x_2$
1	1	1	1	$w_1 + w_2$
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- To get correct labelling we need:
 - $w_1 + w_2 \geq 5$
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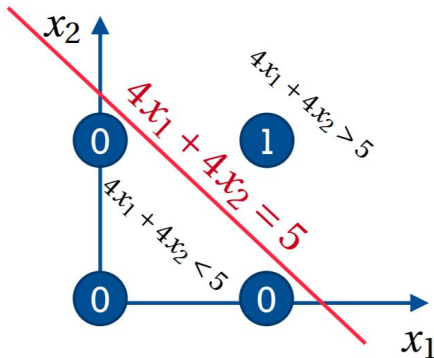
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 - $w_1, w_2 < 5$
- For example: $w_1 = w_2 = 4$

Linear classifier example

E-mail	x_1	x_2	Spam?
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1 = spam

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Training a spam detector

- Suppose now we have 100 features and more e-mails

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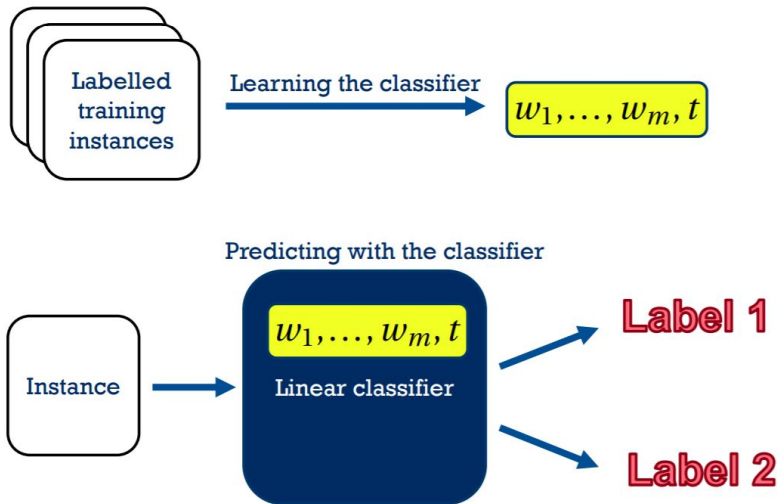
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- We need to personalize!
- Different training datasets for everyone
- Therefore, new set of weights is required for everyone! The process needs to be automated.

Learning a linear classifier and predicting with it

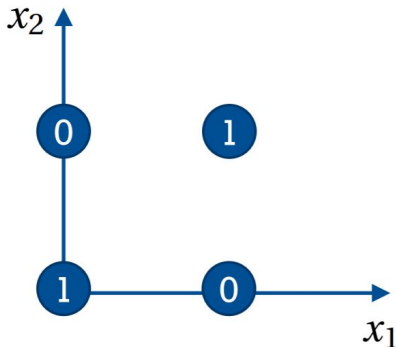


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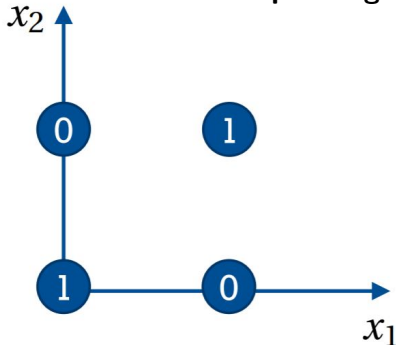
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Seems like we cannot draw a separating line for these two classes.



No linear classifier fits training data?

- Another way of saying it is that *the classes are not linearly separable*
- Possible reasons
 - The data is noisy
 - Linear classifiers are not expressive enough for the given task
- Options
 - Choose a linear classifier which fits as many data as possible
 - Choose a more expressive model type

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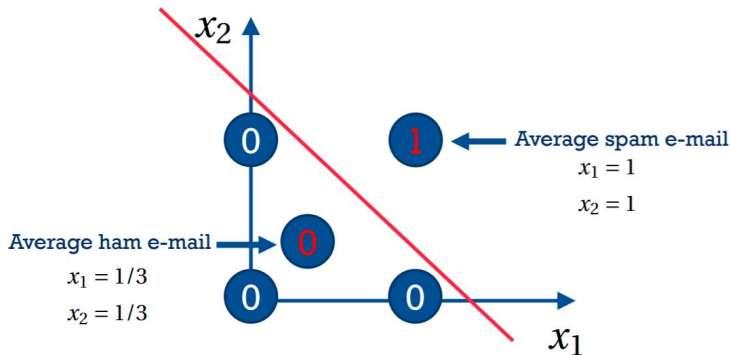
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- Let's compare against average spam and average ham e-mail!

Basic linear classifier

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- Thus, we will predict positive if $\mathbf{x}$ is closer to $\mathbf{p}$ than $\mathbf{n}$:

$$\|\mathbf{x} - \mathbf{n}\| \geq \|\mathbf{x} - \mathbf{p}\|$$

Derivation of weights

$$\sqrt{(\mathbf{x} - \mathbf{n})^T (\mathbf{x} - \mathbf{n})} \geq \sqrt{(\mathbf{x} - \mathbf{p})^T (\mathbf{x} - \mathbf{p})}$$

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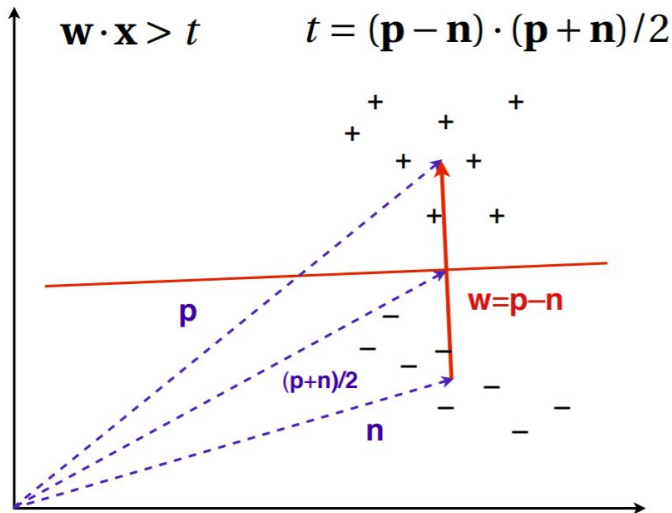
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where $\mathbf{w} = \mathbf{p} - \mathbf{n}$, $t = (\mathbf{p} - \mathbf{n}) \cdot (\mathbf{p} + \mathbf{n})/2 = (\|\mathbf{p}\|^2 - \|\mathbf{n}\|^2)/2$

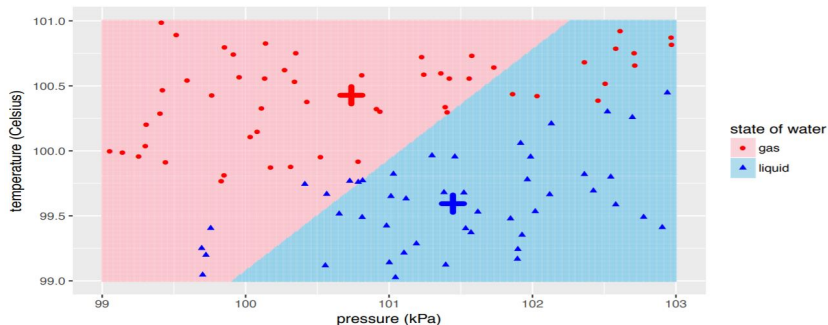
Basic linear classifier



Summary: Basic linear classifier algorithm

- Binary classification method, not widespread
- Learning the model (also called **fitting**):
 - Calculate the centre of mass of positive (training) instances and of negative instances
- Applying the model (also called **predicting**):
 - for a test instance $\mathbf{x}$ predict positive if $\mathbf{w}^T \mathbf{x} > t$
 - otherwise predict negative
 - here $\mathbf{w} = \mathbf{p} - \mathbf{n}$, $t = (\mathbf{p} - \mathbf{n}) \cdot (\mathbf{p} + \mathbf{n})/2$

Problem with basic linear classifier



Main problem

Even though a perfect linear classifier seems to exist, it is not found by the basic linear classifier.

In the future lectures, we will learn about linear classifiers that don't have this type of failure in cases when the label classes are linearly separable.

What have we learned today?

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier